Summary of Data Exploration and Preprocessing

Project Title:

Customer Churn Prediction and Analysis

Key Insights from Data Exploration

1. Understanding the Dataset:
   - Data loaded from a CSV file named 'churn.csv'.
   - Initial inspection suggests it includes customer-related attributes relevant to churn behavior.

2. Visual Exploration:
   - Feature distributions were visualized to detect:
     - Skewed distributions.
     - Outliers (using boxplots).
     - Correlation patterns (using heatmaps).

3. Target Variable:
   - The variable 'Churn' is the primary focus for prediction.
   - Class imbalance may have been considered for handling.

Features in the Dataset

Numerical Features (Scaled & Handled for Outliers):

- - age
  - days_since_last_login
  - avg_time_spent
  - avg_transaction_value
  - avg_frequency_login_days
  - points_in_wallet

Binary Categorical Features (Label Encoded):

- - gender
  - joined_through_referral
  - used_special_discount

- offer_application_preference
- past_complaint

Multi-class Categorical Features (One-Hot Encoded):

- - region_category
  - membership_category
  - medium_of_operation
  - internet_option
  - complaint_status
  - feedback
  - preferred_offer_types

❌ Dropped Feature:

- - security_no (non-informative or identifier column)

Preprocessing Decisions Made

1. Encoding Categorical Variables:
   - Binary features were label-encoded.
   - Multi-class features were one-hot encoded with first category dropped to avoid multicollinearity.

2. Scaling Numerical Features:
   - StandardScaler was applied to normalize numerical features.

3. Handling Skewness:
   - Box-Cox transformation applied where necessary.

4. Missing Values:
   - Numerical features filled with the median.
   - Categorical features filled with the mode.

5. Outlier Treatment:
   - Used IQR method to cap outliers in selected numerical features.

6. Correlation Analysis:
   - Heatmaps likely used to identify and possibly drop highly correlated features.

☑ Conclusion

This thorough preprocessing pipeline ensures the data is clean, numerically stable, and appropriately encoded for model training.
Such preparation significantly boosts the accuracy and reliability of churn prediction models.